SAPONIFICATION 23

Saponification (literally 'making soap') is an important industrial process.

Saponification is the process of using basic (alkaline) solutions to convert fats and oils (lipids) into glycerol and salts of fatty acids.

1 State one physical difference between a fat and an oil.

Fats and oils are types of **esters**. Figure 1 shows the general structure of an ester. R_1 and R_2 are different alkyl groups. Esters are formed by the reaction between an alkanoic (carboxylic) acid and an alkanol (alcohol):

Acid + alkanol ⇌ ester + water

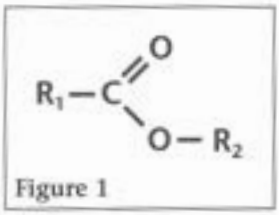


Figure 1

2 What is the significance of the symbol '⇌' in the equation above?

Fats and oils tend to be long chain triglycerides (*tri*, meaning three). Figure 2 shows the structures of glycerol and a typical triglyceride. The R is a long chain hydrocarbon group.

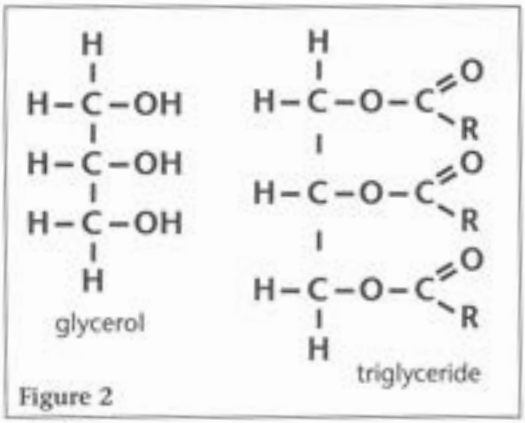


Figure 2

3 Give the systematic (IUPAC) name for glycerol.

4 Glyceryl tristearate is a typical animal fat. Using the structure shown in figure 2 and given that the R (stearate) group for glyceryl tristearate has 17 carbon atoms, draw the structure for this compound. Assume that the hydrocarbon group is saturated.

One version of the discovery of soap is that animal fats from meat being cooked over a wood fire dripped into the ashes. The ashes contained alkaline potassium carbonate that reacted with the fats to form an impure form of soap. The people who raked out the ashes then washed their hands in water found that the mixture cleaned their hands better than water alone did. Around 200 years ago, soap was made by boiling animal fats with alkaline solutions. A similar process is used today but generally vegetable oils are used and there are extra steps to ensure the purity of the finished product.

A triglyceride is heated with sodium hydroxide solution forming glycerol and a sodium carboxylate (soap). Figure 3 shows an outline of this process. Glyceryl tripalmitate (a fat derived from palms) is commonly used as a raw material. It forms sodium palmitate.

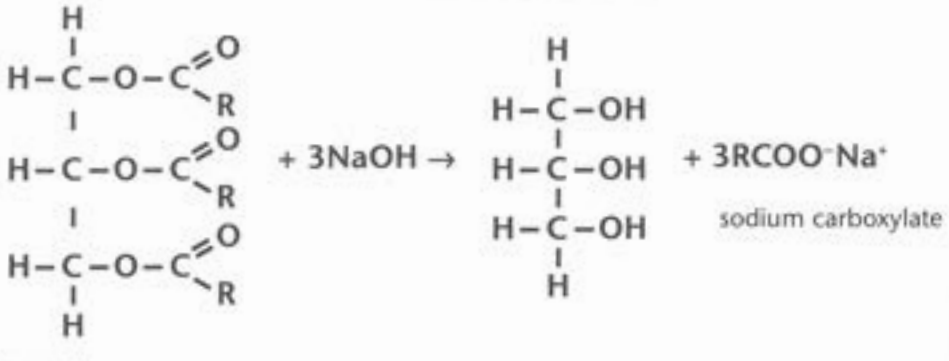


Figure 3

5 Using the structure shown in figure 3 and given that the R (palmitate) group for sodium palmitate has 15 carbon atoms, draw the structure for this compound. Assume that the hydrocarbon group is saturated.

The industrial production of soap can involve the breaking down of the triglyceride into fatty acids before it is treated with sodium hydroxide. Fatty acids do not occur naturally as separate compounds, but only bound with glycerol. The use of potassium hydroxide instead of sodium hydroxide gives physically softer soaps.

6 Name two other substances that manufacturers might add to soaps.

7 *The sodium and potassium salts of fatty acids*. What does this define?

8 *Tallow* is another name for animal fat. True or false?

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